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Plant Diseases Identification by Machine Learning Techniques: A Literature Review

Aruna Mittal | Ashok Kumar | Shelendra Pal | Computer Science, Agricultural and Food Sciences ·

2025 5th International Conference on Advancement… · 21 November 2025

TLDR This work emphasizes a method to get excellent, full-color photos of plant lesion leaves to improve classification accuracy in **deep learning** models, and offers insightful analysis of present trends and recommendations for future study paths to overcome present difficulties in plant Disease identification by **artificial intelligence**-driven methods. Expand

IEEE | Save | Cite

Deep Learning for B5G Open Radio Access Network: Evolution, Survey, Case Studies, and Challenges

Bouziane Brik | Karim Boutiba | A. Ksentini | Computer Science, Engineering ·

IEEE Open Journal of the Communications Society · 2022

TLDR This work describes how the main steps of deployed DL models in O-RAN can be automated, to ensure stable performance of these models, and identifies key technical challenges, open issues, and future research directions related to the **Artificial Intelligence** (AI)-enabled O- RAN architecture. Expand

117 | PDF | IEEE | Save | Cite

Deep Learning Sensor Fusion in Plant Water Stress Assessment: A Comprehensive Review

M. Kamarudin | Z. Ismail | N. Saidi | Environmental Science, Computer Science · Applied Sciences ·

4 February 2021

TLDR This paper extensively **reviews** the state-of-the-art methods for plant water stress assessment that utilized the **deep learning** sensor fusion approach in their application, together with future prospects and challenges of the application domain. Expand

48 | [PDF] | PDF | Save | Cite

A Comprehensive Survey on Machine Learning and Deep Learning Techniques for Diabetes Prediction

Student | Medicine, Computer Science

TLDR The research shows that ensemble and hybrid models tend to perform better than single classifiers, with some models having a prediction accuracy of up to 100% and some models having a prediction accuracy of up to 100%. Expand

1 | PDF | Save | Cite

Spatiotemporal Patterns of Aboveground Carbon Storage in Hainan Mangroves Based on Machine Learning and Multi-Source Remote Sensing Data

Zhikuan Liu | Zhaode Yin | +4 authors | Zixuan Qiu | Environmental Science, Computer Science ·

Forests · 19 January 2026

As an essential blue carbon ecosystem, mangroves play a vital role in coastal protection, biodiversity conservation, and climate regulation. However, their complex and variable growth environments… Expand

Publisher | Save | Cite

Deep Learning-Driven Student Performance Analysis: Detecting Anomalies and Predicting Academic Success

Muhammad Nadeem Gul · Muhammad Arif · Sidra Gulzar · Ghulshan Naveed · Waseem Abbasi

Computer Science, Education · Inverge Journal of Social Sciences · 28 March 2025

TLDR This research makes a substantial contribution to the growing field of Educational Data Mining by proposing a robust **deep learning** framework that serves as both a predictive tool and a baseline for future studies in student performance analysis, while also addressing critical challenges in model interpretability and implementation scalability within real-world educational settings. Expand

3 · Publisher · Save · Cite

Multidimensional Advances in Wildfire Behavior Prediction: Parameter Construction, Model Evolution and Technique Integration

Hai-Hui Wang · Kai Zhang · Shamima Aktar · Zezhen Wu

Environmental Science, Computer Science · Fire · 16 October 2025

TLDR The status and future development trends of **wildfire** behavior modeling and prediction technologies are **reviewed**, providing a comprehensive overview of the evolution of models from empirical to physical and then to data-driven approaches, emphasizing the integration of multidisciplinary **techniques** such as **machine learning** and **deep learning**. Expand

Publisher · Save · Cite

Aerial Imagery Pile burn detection using Deep Learning: the FLAME dataset

Alireza Shamsoshoara · F. Afghah · Abolfazl Razi · Liming Zheng · Peter Z Ful'e · Erik Blasch

Environmental Science, Computer Science · Comput. Networks · 28 December 2020

TLDR A **fire** image dataset collected by drones during a prescribed burning piled detritus in an Arizona pine **forest** is provided and a **deep learning** method is designed based on the U-Net up-Sampling and down-sampling approach to extract a **fire** mask from the video frames. Expand

303 · [PDF] · arXiv · Save · Cite

An Extensive Review of Recent Development of Solar Power Prediction Models

C. Madan · Environmental Science, Engineering ·

Journal of Computational Mechanics Power System… · 2025

TLDR A literature **review** was performed that concentrated on the most important papers that were published in IEEE, Google Scholar, Science Direct, and Springer journals between 2015 and 2024, and examined methods that take into consideration convergence time, tracking efficiency, and error metrics. Expand

PDF · Publisher · Save · Cite

Starting an eResearch revolution with deep learning

Brent Martin · Aleksandra Pawlik · Environmental Science, Computer Science · 26 February 2021

TLDR A **deep learning** model can classify beech pollen species from images to over 80% accuracy, a task considered too difficult for even specialist humans, and is being achieved through a collaboration between the Informatics research team and the wider scientific cohort at MWLR. Expand

Publisher · Save · Cite

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